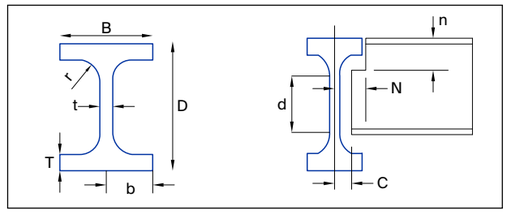
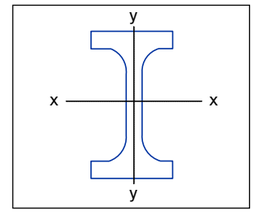
2.4
Metric Beams & Columns (MB & MC)



Dimensions	JIS G 3192
Specification	EN10025/ ASTM A36/ ASTM A572/ JIS G 3101
Size Range	MB: 100mm x 50mm to 900mm x 300mm MC: 100mm x 100mm to 400mm x 400mm



Metric Units

Designation		Depth of Section	Width of Section	Thickness		Root Radius	Depth between Fillets	Area of Section	Ratios for Local Buckling		Dimensions for Detailing		
				Flange	Web				Flange	Web	End clearance	Notch	
Size	Mass per Metre	D	B	T	t	r	d	A	b/T	d/t	C	N	n
in (mm)	kg/m	mm	mm	mm	mm	mm	mm	cm²			mm	mm	mm
250x125	25.1	248	124	8	5	8	216	32.0	7.75	43.2	5	70	16
	25.7	248	124	8	5	12	208	32.7	7.75	41.6	5	70	20
	29.0	250	125	9	6	8	216	37.0	6.94	36.0	5	70	17
	29.6	250	125	9	6	12	208	37.7	6.94	34.7	5	70	21
250x175	43.6	244	175	11	7	13	196	55.5	7.95	28.0	6	94	24
	44.1	244	175	11	7	16	190	56.2	7.95	27.1	6	94	27
	51.6	248	176	13	8	16	190	65.7	6.77	23.8	6	94	29
	59.1	252	177	15	9	16	190	75.3	5.90	21.1	7	94	31
250x250	66.5	248	249	13	8	16	190	84.7	9.58	23.8	6	131	29
	71.8	250	250	14	9	13	196	91.4	8.93	21.8	7	131	27
	72.4	250	250	14	9	16	190	92.2	8.93	21.1	7	131	30
	81.6	250	255	14	14	13	196	104	9.11	14.0	9	131	27
	98.1	260	253	19	12	16	190	125	6.66	15.8	8	131	35
300x150	25.0	294	148	6	4.5	16	250	32.6	12.3	55.6	4	82	22
	32.0	298	149	8	5.5	13	256	40.8	9.31	46.5	5	82	21
	36.7	300	150	9	6.5	13	256	46.8	8.33	39.4	5	82	22
	41.4	304	150	11	6.5	13	256	52.8	6.82	39.4	5	82	24
	46.2	306	151	12	7.5	13	256	58.8	6.29	34.1	6	82	25
	69.0	318	154	18	11	13	256	87.9	4.28	23.3	8	82	31
300x200	48.3	290	199	10	7	18	234	61.5	9.95	33.4	6	106	28
	55.8	294	200	12	8	13	244	71.1	8.33	30.5	6	106	25
	56.8	294	200	12	8	18	234	72.4	8.33	29.3	6	106	30
	65.4	298	201	14	9	18	234	83.4	7.18	26.0	7	106	32
	77.3	304	202	17	10	18	234	98.5	5.94	23.4	7	106	35
300x300	87.0	298	299	14	9	18	234	111	10.7	26.0	7	155	32
	93.0	300	300	15	10	13	244	118	10.0	24.4	7	155	28
	94.0	300	300	15	10	18	234	120	10.0	23.4	7	155	33
	105	300	305	15	15	13	244	133	10.2	16.3	10	155	28
	106	304	301	17	11	18	234	135	8.85	21.3	8	155	35
	125	310	303	20	13	18	234	159	7.58	18.0	9	155	38
	130	310	305	20	15	18	234	165	7.63	15.6	10	155	38
	147	312	310	21	20	18	234	187	7.38	11.7	12	155	39

Surface Area per Metre	Second Moment of Area		Radius of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant
Per metre	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	u	x	H	J
m²	cm⁴	cm⁴	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
0.968	3450	255	10.4	2.82	278	41.1	312	63.2	0.884	32.0	0.037	5.78
0.961	3537	255	10.4	2.79	285	41.1	319	63.6	0.886	29.9	0.037	6.74
0.974	3965	294	10.4	2.82	317	47.0	358	72.7	0.880	28.4	0.043	8.51
0.967	4052	294	10.4	2.79	324	47.0	366	73.1	0.881	26.9	0.043	9.68
1.15	6037	984	10.4	4.21	495	112	550	172	0.883	21.3	0.134	21.2
1.15	6122	985	10.4	4.19	502	113	558	173	0.884	20.5	0.134	23.2
1.16	7308	1184	10.5	4.24	589	135	660	207	0.884	18.0	0.163	36.1
1.17	8541	1390	10.7	4.30	678	157	764	241	0.885	16.0	0.195	53.2
1.45	9931	3348	10.8	6.29	801	269	883	408	0.850	17.9	0.462	46.7
1.46	10750	3648	10.8	6.32	860	292	953	443	0.847	17.1	0.508	55.8
1.45	10830	3649	10.8	6.29	867	292	960	444	0.848	16.7	0.508	58.7
1.47	11400	3876	10.5	6.11	912	304	1031	467	0.826	15.7	0.540	75.4
1.48	15340	5134	11.1	6.41	1180	406	1330	618	0.849	12.8	0.745	141
1.14	4940	326	12.3	3.16	336	44.0	375	68.4	0.878	39.7	0.068	5.51
1.16	6318	442	12.4	3.29	424	59.4	475	91.8	0.880	35.6	0.093	8.65
1.16	7210	508	12.4	3.29	481	67.7	542	105	0.876	32.0	0.107	12.4
1.17	8578	620	12.7	3.43	564	82.7	633	128	0.888	27.8	0.133	18.8
1.18	9514	691	12.7	3.43	622	91.5	702	142	0.884	25.6	0.149	24.9
1.21	14820	1100	13.0	3.54	932	143	1070	223	0.884	17.9	0.248	79.0
1.33	9431	1317	12.4	4.63	650	132	721	203	0.882	26.0	0.258	22.8
1.35	11110	1602	12.5	4.75	756	160	842	245	0.882	24.0	0.319	31.4
1.34	11340	1604	12.5	4.71	771	160	859	247	0.883	22.7	0.319	35.8
1.35	13310	1900	12.6	4.77	893	189	1000	291	0.884	20.1	0.383	53.4
1.37	16280	2341	12.9	4.88	1071	232	1204	356	0.888	17.2	0.482	87.8
1.74	18850	6242	13.0	7.51	1265	418	1389	634	0.850	20.0	1.26	71.3
1.76	20190	6753	13.1	7.55	1346	450	1484	683	0.847	19.4	1.37	82.1
1.75	20410	6756	13.1	7.51	1361	450	1501	684	0.848	18.8	1.37	88.1
1.77	21310	7103	12.6	7.30	1421	466	1596	714	0.828	17.9	1.44	109
1.76	23380	7733	13.2	7.57	1538	514	1705	781	0.849	16.9	1.59	125
1.78	28130	9282	13.3	7.64	1815	613	2031	932	0.849	14.6	1.95	200
1.78	28630	9470	13.2	7.57	1847	621	2079	949	0.842	14.4	1.99	215
1.79	31370	10450	13.0	7.48	2011	674	2295	1040	0.830	13.0	2.21	298